NuLEDCtrl USB Protocol

Application Note for 32-bit NuMicro® Family

Document Information

Abstract	This document introduces the USB protocol from PC Tool NuLEDCtrl to control NuMaker-Lighting-ARGB.
Apply to	NuMaker-Lighting-ARGB.

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1 Overview

NuLEDCtrl uses USB interface to connect with NuMaker-Lighting-ARGB. The USB Class is HID and VID is **0x0416**, PID is **0x5020**.

There are 6 kinds of command:

- General Command
  This kind of command is used to get FW version, get FW checksum, start ISP process, transmit music volume.
- LED Strip Set Command
  This kind of command is used to set LED numbers on strip, RG swap, direction, and Independent/Sync mode.
- LED Mode Set Command
  This kind of command is used to set LED lighting mode, LED color (R, G, B), brightness and speed.

2 General Command

Uses 0x01 as Report ID.

2.1 Get FW Version (0xA0)

Get FW version, which includes major and minor bytes.

2.1.1 Host ask Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Command	Reserved					
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

• **Report ID:** 0x01 • **Command:** 0xA0 • **Reserved:** 0x00		

2.1.2 Device reply Host

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Command	Major	Minor	Reserved			
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

• **Report ID:** 0x01 • **Command:** 0x5A • **Major:** Based on FW version • **Minor:** Based on FW version • **Reserved:** 0x00		

2.2 Get FW Checksum (0xA1)

Get FW checksum.

2.2.1 Host ask Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Command	Reserved					
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

- **Report ID:**
  0x01
- **Command:**
  0xA1
- **Reserved:**
  0x00

2.2.2 Device reply Host

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Command	Checksum		Reserved			
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

- **Report ID:**
  0x01
- **Command:**
  0x5A
- **Checksum:**
  Byte 3 is Low Byte
  Byte 4 is High Byte
- **Reserved:**
  0x00

2.3 Start ISP Process (0xB0)

Ask to enter ISP process. LED Dongle will restart and boot from LDROM to continue ISP Process based on Nuvoton USB ISP.

2.3.1 Host ask Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Command	Reserved					
Byte 9	Byte 10	Byte 11	Byte 412	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

• **Report ID:** 0x01 • **Command:** 0xB0 • **Reserved:** 0x00		

2.4 Set Music Volume (0xC0)

Set music volume and action to Device.

2.4.1 Host set Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Command	Music Volume			Action	Reserved	
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

• **Report ID:** 0x01 • **Command:** 0xC0 • **Music Volume:** Byte 3 is Main Volume Byte 4 is Left Volume Byte 5 is Right Volume	• **Action:** 0x01 = POP Whole strip color is based on volume. Blue (Min.) <=> Red (Max.) 0x02 = JAZZ Lighten LED number based on volume. 0x03 = Mixed Mixed POP and JAZZ mode. Lighten LED number and color based on volume.	• **Reserved:** 0x00

3 LED Strip Set Command

Uses 0x02 as Report ID.

3.1 Set LED Strip Configuration

Set LED numbers on strip, RG swap, direction, and Independent/Sync Mode.

3.1.1 Host set Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Area		LED Numbers		RG Swap	Direction	AP Sync
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Reserved							

- **Report ID:**
  0x02

- **Area:**
  Byte 2 is Low Byte
  Byte 3 is High Byte
  Bit0 = Area 1 (Strip A)
  Bit1 = Area 2 (Strip B)
  Bit2 = Area 3 (Strip C)
  Bit3 = Area 4 (Strip D)

- **LED Numbers:**
  Byte 4 is Low Byte
  Byte 5 is High Byte
  1 ~ 10 (Max value depends on value NUM_STRIP_LED_EVB defined in device.h.)

- **RG Swap:**
  0x00 = GRB Type
  0x01 = RGB Type

- **Direction:**
  0x00 = Forward
  0x01 = Backward

- **AP Sync:**
  0x00 = Disable
  0x01 = Enable

- **Reserved:**
  0x00

- **Reserved:**
  0x00

4 LED Mode Set Command

Uses 0x03 as Report ID.

4.1 Set LED Lighting Mode Configuration

Set LED lighting mode, LED color (R, G, B), brightness and speed.

4.1.1 Host set Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Area		Lighting Mode	Color R	Color G	Color B	Brightness
Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16
Speed	Reserved						

- **Report ID:**
  0x03

- **Area:**
  Byte 2 is Low Byte
  Byte 3 is High Byte
  Bit0 = Area 1 (Strip A)
  Bit1 = Area 2 (Strip B)
  Bit2 = Area 3 (Strip C)
  Bit3 = Area 4 (Strip D)

- **Lighting Mode:**
  0x00: LED_OFF
  0x01: LED_STATIC
  0x02: LED_BREATHING
  0x03: LED_STROBE
  0x04: LED_CYCLING
  0x05: LED_RANDOM
  0x06: LED_MUSIC
  0x07: LED_WAVE
  0x08: LED_SPRING
  0x09: LED_WATER
  0x0A: LED_RAINBOW

- **Color R:**
  0x00 ~ 0xFF

- **Color G:**
  0x00 ~ 0xFF

- **Color B:**
  0x00 ~ 0xFF

- **Brightness:**
  0x00 (Min.) ~ 0xFF (Max.)

- **Speed**
  0x00 (Fast) ~ 0xFF (Slow)

- **Reserved:**
  0x00

5 LED AP Sync Command

Uses 0x04 as Report ID.

Uses set feature report to transmit.

5.1 Set LED Sync Color

Set each LED color from AP Tool.

5.1.1 Host set Device

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Report ID	Area		LED 1 Color			LED 2 Color	
Byte 9	Byte 10	Byte 11	Byte 12	.....	Byte 433	Byte 434	Byte 435
LED 2 Color	LED 3 Color			.....	LED 144 Color		

- **Report ID:**
  0x04

- **Area:**
  Byte 2 is Low Byte
  Byte 3 is High Byte
  Bit0 = Area 1 (Strip A)
  Bit1 = Area 2 (Strip B)
  Bit2 = Area 3 (Strip C)
  Bit3 = Area 4 (Strip D)

- **LED n Color:**
  Byte (n*3 + 1) is Color R
  Byte (n*3 + 2) is Color G
  Byte (n*3 + 3) is Color B

Revision History

Date	Revision	Description
2024.06.28	1.00	1. Initial version.

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